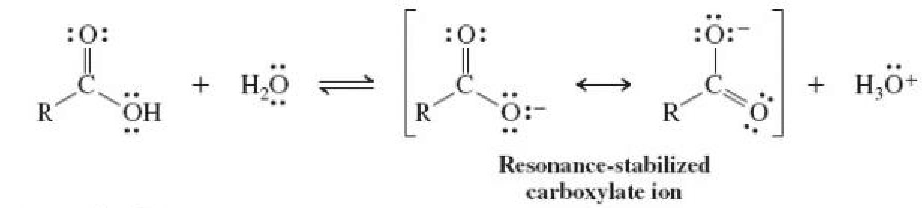
New Reactions

1. **Acidity of Carboxylic Acids ([Section 19-4](#))**



$K_a = 10^{-4}$–10^{-5}, $pK_a \approx 4$–5

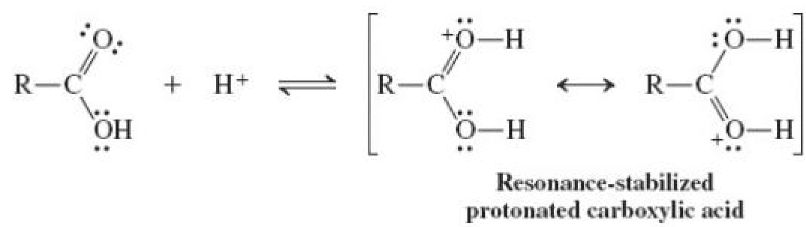
Salt formation

RCOOH+NaOH→RCOO-Na++H2OAlso with Na2CO3,NaHCO3

$$RCOOH + NaOH \rightarrow RCOO^{-}Na^{+} + H_2O$$

$$\mathbf{Also\ with\ Na_2\,CO_3, NaHCO_3}$$

2. **Basicity of Carboxylic Acids ([Section 19-4](#))**



Preparation of Carboxylic Acids

3. **Oxidation of Primary Alcohols and Aldehydes ([Section 19-6](#))**

RCH2OH→Oxidizing agentRCOOHOxidizing agents: aqueous CrO3,KMnO4,HNO3RCHO→Oxidizing agentRCOOHOxidizing agents: aqueous CrO3,KMnO4,Ag+,H2O2,HNO3

$$RCH_2OH \xrightarrow{\text{Oxidizing agent}} RCOOH$$

$$\mathbf{Oxidizing\ agents{:}\ aqueous\ CrO_3, KMnO_4, HNO_3}$$

$$RCHO \xrightarrow{\text{Oxidizing agent}} RCOOH$$

$$\mathbf{Oxidizing\ agents{:}\ aqueous\ CrO_3, KMnO_4, Ag^{+}, H_2O_2, HNO_3}$$

4. **Carbonation of Organometallic Reagents ([Section 19-6](#))**

RMgX+CO2→THFRCOO-
+MgX→H+,H2ORCOOHRLi+CO2→THFRCOO-
Li+→H+,H2ORCOOH

$$RMgX + CO_2 \xrightarrow{THF} RCOO^{-+}MgX \xrightarrow{H^+, H_2O} RCOOH$$
$$RLi + CO_2 \xrightarrow{THF} RCOO^-Li^+ \xrightarrow{H^+, H_2O} RCOOH$$

5. **Hydrolysis of Nitriles ([Section 19-6](#))**

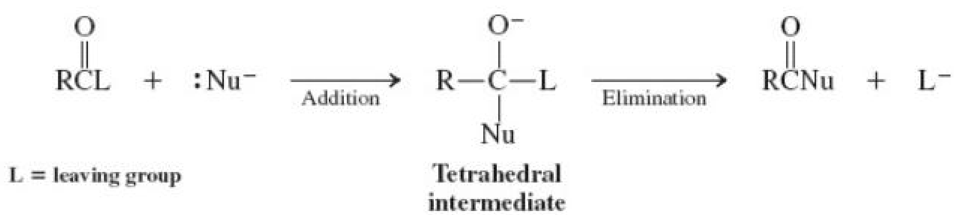
RC≡N→H2O,Δ,H+ or HO−RCOOH+NH3orNH4+

$$RC \equiv N \xrightarrow{H_2O, \Delta, H^+ \text{ or } HO^-} RCOOH + NH_3 \text{ or } NH_4{}^+$$

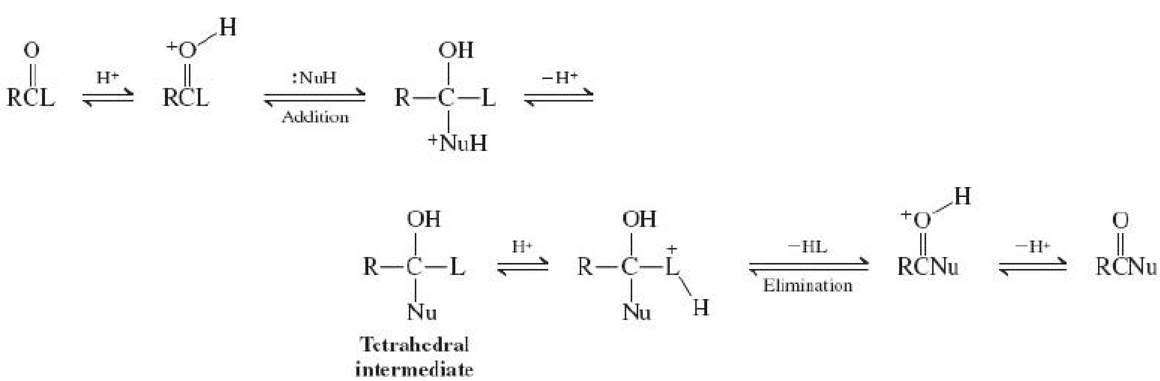
Reactions of Carboxylic Acids

6. **Nucleophilic Attack at the Carbonyl Group ([Section 19-7](#))**

Base-catalyzed addition–elimination



Acid-catalyzed addition–elimination



Derivatives of Carboxylic Acids

7. **Acyl Halides (Section 19-8)**

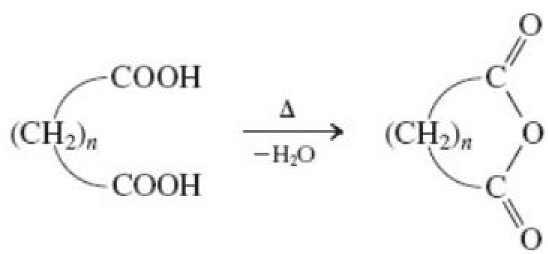
$$\mathrm{RCOOH} + \mathrm{SOCl_2} \longrightarrow \underset{\textbf{Acyl chloride}}{\mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}Cl}} + \mathrm{SO_2} + \mathrm{HCl}$$

$$3\,\mathrm{RCOOH} + \mathrm{PBr_3} \longrightarrow 3\,\underset{\textbf{Acyl bromide}}{\mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}Br}} + \mathrm{H_3PO_3}$$

8. **Carboxylic Anhydrides (Section 19-8)**

$$\mathrm{RCOOH} + \mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}Cl} \longrightarrow \underset{\textbf{Anhydride}}{\mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}O\overset{\overset{\displaystyle O}{\|}}{C}R}} + \mathrm{HCl}$$

Cyclic anhydrides



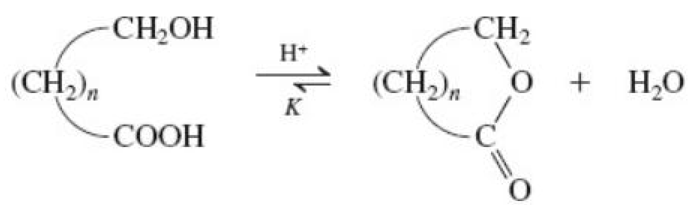
Best for five- or six-membered rings

9. **Esters (Section 19-9)**

Acid-catalyzed esterification

$$\mathrm{RCO_2H} + \mathrm{R'OH} \underset{K \approx 1}{\overset{\mathrm{H^+}}{\rightleftharpoons}} \mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}OR'} + \mathrm{H_2O}$$

Cyclic esters (lactones)



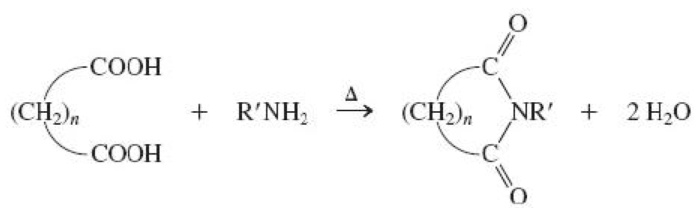
Lactone

$K > 1$ for five- and six-membered rings

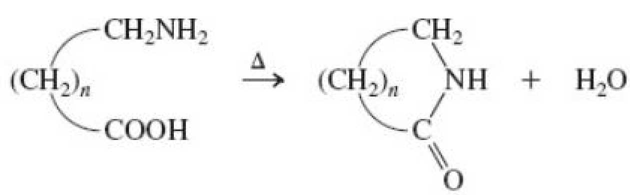
10. Carboxylic Amides (Section 19-10)

$$\mathrm{RCOOH} + \mathrm{R'NH_2} \longrightarrow \mathrm{RCOO^-} + \mathrm{R'NH_3^+} \xrightarrow{\Delta} \mathrm{R\overset{O}{\overset{\|}{C}}NHR'} + \mathrm{H_2O}$$

Imides



Cyclic amides (lactams)



Lactam

11. Reduction with Lithium Aluminum Hydride (Section 19-11)

RCOOH→1.LiAlH4,(CH3CH2)2O2.H+,H2ORCH2OH

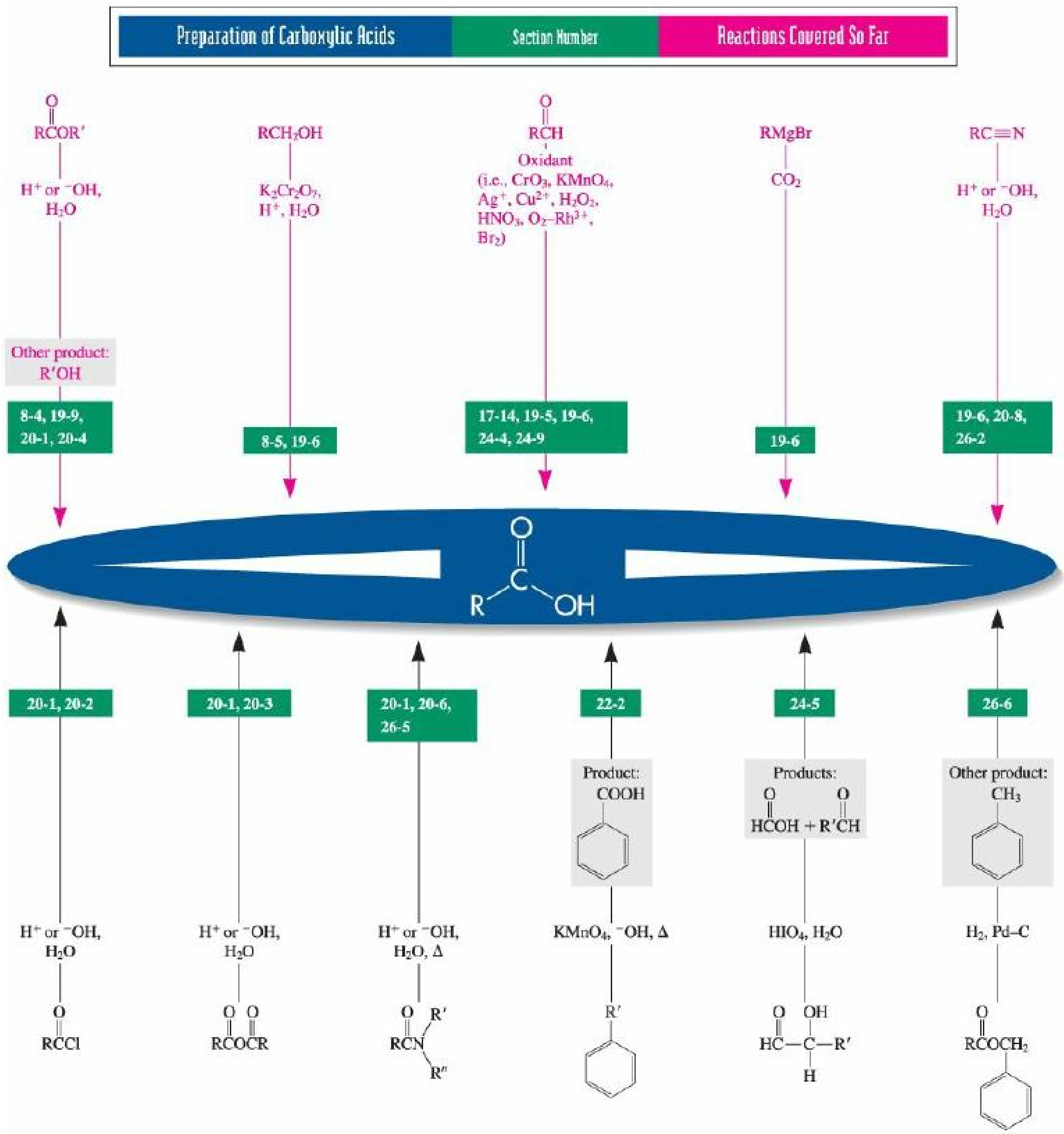
$$\mathrm{RCOOH} \xrightarrow[2.\mathrm{H^+,H_2O}]{1.\mathrm{LiAlH_4,(CH_3CH_2)_2O}} \mathrm{RCH_2OH}$$

12. Bromination: Hell-Volhard-Zelinsky Reaction (Section 19-12)

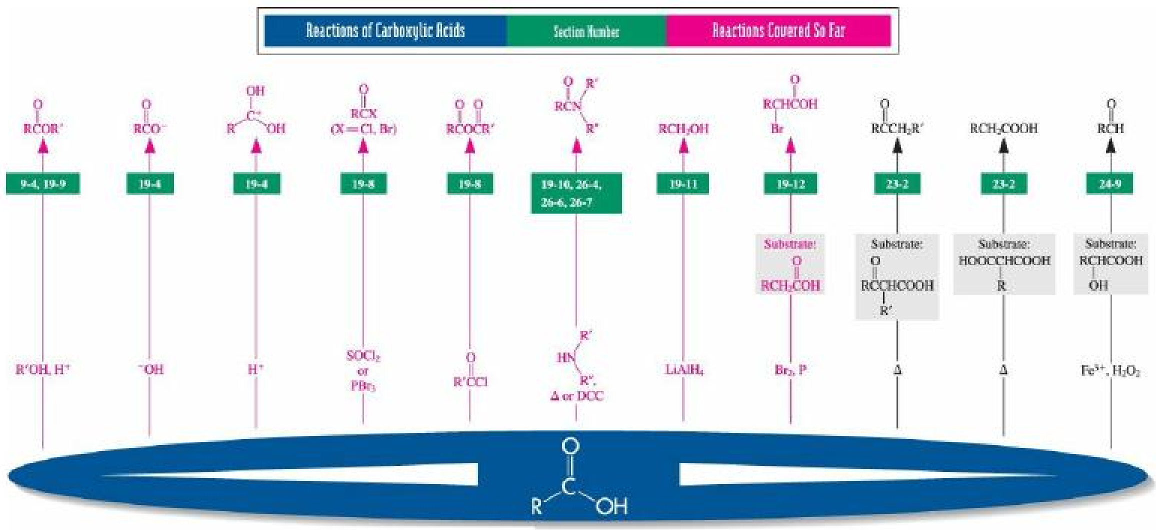
$$\mathrm{RCH_2COOH} \xrightarrow{\mathrm{Br_2,\ trace\ P}} \mathrm{R\underset{}{\overset{Br}{\overset{|}{C}}}HCOOH}$$

Important Concepts

1. **Carboxylic acids** are named as **alkanoic acids.** The carbonyl carbon is numbered 1 in the longest chain incorporating the carboxy group. Dicarboxylic acids are called **alkanedioic acids.** Cyclic and aromatic systems are called **cycloalkanecarboxylic** and **benzoic acids,** respectively. In these systems the ring carbon bearing the carboxy group is assigned the number 1.
2. The **carboxy group** is approximately **trigonal planar.** Except in very dilute solution, carboxylic acids form dimers by hydrogen bonding.
3. The carboxylic acid **proton chemical shift** is variable but relatively **high** (δ=10-13),$(\delta = 10 - 13)$, because of hydrogen bonding. The **carbonyl carbon** is also relatively **deshielded** but not as much as in aldehydes and ketones, because of the resonance contribution of the hydroxy group. The carboxy function shows two important infrared bands, one at about 1710 $cm^{-1}$$1710\ \mathrm{cm}^{-1}$ for the C=OCâ O bond and a very broad band between 2500 and 3300 $cm^{-1}$$2500 \text{ and } 3300\ \mathrm{cm}^{-1}$ for the O−H$\mathrm{O-H}$ group. The mass spectrum of carboxylic acids shows facile fragmentation in three ways.
4. The carbonyl group in carboxylic acids undergoes nucleophilic displacement via the **addition–elimination** pathway. Addition of a nucleophile gives an unstable **tetrahedral intermediate** that decomposes by elimination of the hydroxy group to give a **carboxylic acid derivative.**
5. **Lithium aluminum hydride** is a strong enough nucleophile to add to the carbonyl group of carboxylate ions. This process allows the reduction of carboxylic acids to **primary alcohols.**



Vollhardt/Schore, *Organic Chemistry*, 8e, © 2018, W. H. Freeman and Company


Reactions of Carboxylic Acids
Section Number
Reactions Covered So Far
RCOR′
RCO⁻
RCX (X = Cl, Br)
RCOCR′
RCH₂OH
RCHCOH Br
RCCH₂R′
RCH₂COOH
RCH
9-4, 19-9
19-4
19-4
19-8
19-8
19-10, 26-4, 26-6, 26-7
19-11
19-12
23-2
23-2
24-9
Substrate: RCH₂COH
Substrate: RCCHCOOH R′
Substrate: HOOCCHCOOH R
Substrate: RCHCOOH OH
R′OH, H⁺
⁻OH
H⁺
SOCl₂ or PBr₃
R′CCl
Δ or DCC
LiAlH₄
Br₂, P
Δ
Δ
Fe³⁺, H₂O₂